BTEUP EXAM NOTES | HINGLISH EDITION

Information Security

Unit 2: Network Protocol and Device Weaknesses and Security Solutions

In notes ke baare me

Part A me har topic simple Hinglish me real-life example ke saath samjhaya hai, taaki beginner bhi zero se samajh sake.
Part B me important exam questions ke poore 5-marks wale answers Hinglish me diye hain.
Last me: quick revision sheet, common mistakes aur self-test.
Exam me answer English me likhna hai to English Edition PDF use karo.

Syllabus covered (Unit 2)

Understanding security weaknesses in popular networking protocols: IP, TCP, UDP, RIP, OSPF, HTTP, SMTP, etc.; security weaknesses in common networking devices: Hub, switch, router, WiFi; Security solutions to mitigate security risk of networking protocols (IPSec, HTTPS, etc) and devices (VLAN, VPN, Ingress Filtering, etc).

Unit Overview aur Contents

Simple idea: Networks ko *fast aur easy* banane ke liye design kiya gaya tha, *secure* banane ke liye nahi. Purane protocols aur devices sabhi par bharosa karte hain, isliye attackers unka galat fayda uthate hain. Is unit me 3 parts hain: (1) protocols ki **weaknesses**, (2) devices ki **weaknesses**, (3) inhe fix karne wale **solutions**.

Topic	Exam importance	Notes me kahan
IP, TCP, UDP ki weaknesses	Very High	Part A: Topic 2 | Part B: Q1, Q11
RIP, OSPF ki weaknesses	High	Part A: Topic 2 | Part B: Q2
HTTP, SMTP ki weaknesses	Very High	Part A: Topic 2 | Part B: Q3
Hub, Switch, Router, WiFi ki weaknesses	Very High	Part A: Topic 3 | Part B: Q4, Q5, Q12
IPSec, HTTPS	Very High	Part A: Topic 4 | Part B: Q6, Q7
VLAN, VPN, Ingress Filtering	Very High	Part A: Topic 4 | Part B: Q8, Q9, Q10

Contents

Part A: Easy Explanation	Part B: Important Questions aur Answers
Topic 1: Common network attacks Topic 2: Protocol weaknesses Topic 3: Device weaknesses Topic 4: Security solutions	Q1 IP, TCP, UDP weaknesses (Very Likely) Q2 RIP and OSPF weaknesses (Likely) Q3 HTTP and SMTP weaknesses (Very Likely) Q4 Hub, Switch, Router weaknesses (Very Likely) Q5 WiFi weaknesses and security (Very Likely) Q6 IPSec (Very Likely) Q7 HTTPS (Likely) Q8 VLAN (Likely) Q9 VPN (Likely) Q10 Ingress Filtering (Likely) Q11 SYN Flood attack (Possible) Q12 Hub vs Switch, MAC flooding, ARP spoofing (Possible)

Is unit ko kaise padhein (time: 1 din)

1) Part A ek baar padh kar samjho. 2) Part B ke answers diagram ke saath copy me likh kar practice karo. 3) Quick Revision Sheet se revise karo. 4) Self-Test bina dekhe karo.

PART A: Easy Explanation (Hinglish)

Topic 1: Common Network Attacks (basics first)

Real-life example

Network ek sadak jaisa hai. Koi bhi sadak par khade hokar dekh sakta hai ki kya guzar raha hai, ya kisi aur ka naam leke dhokha de sakta hai. Protocol weaknesses samajhne se pehle ye common attacks jaan lo.

Attack	Matlab	Simple example
Sniffing	Network par jaata hua data capture karna	Plain text me bheja password padh lena
Spoofing	Fake pehchan banana (IP, MAC, email)	Fake source IP wala packet bhejna
DoS / DDoS	Server par itna load daalna ki real users ko service na mile	SYN flood, UDP flood
Man-in-the-Middle (MITM)	Attacker do parties ke beech baith kar data padhta ya badalta hai	Fake WiFi hotspot
Session Hijacking	Kisi ka chalu session apne haath me le lena	Login cookie chura lena

Topic 2: Security Weaknesses in Networking Protocols

Protocols weak kyun hain?

Zyada tar protocols (IP, TCP, UDP, HTTP, SMTP) 1970-80 ke time bane the jab internet chhota tha aur sab par bharosa tha. Inme default me **authentication, encryption aur integrity check nahi** hota.

2(a) IP (Internet Protocol)

Real-life example

Post ke letter par sender ka address sender khud likhta hai. Koi bhi galat address likh sakta hai. IP bhi aisa hi hai.

- **Authentication nahi:** source IP address fake kiya ja sakta hai (**IP spoofing**).
- **Encryption nahi:** packets clear text me jaate hain, isliye **sniff** ho sakte hain.
- **Fragmentation attacks:** toote ya overlapping fragments purane systems ko crash kar dete hain (Teardrop, Ping of Death).
- **ICMP abuse:** Smurf attack (fake source ke saath broadcast ping) aur ping flood se DoS hota hai.
- **Fix:** IPSec, ingress/egress filtering, firewall, directed broadcast band karna.

2(b) TCP (Transmission Control Protocol)

Real-life example

TCP phone call jaisa hai: pehle tum 'hello' bolte ho, samne wala 'hello' bolta hai, phir tum confirm karte ho. Fraud wala fake numbers se baar baar 'hello' bol kar dukaan ka phone hamesha busy rakh sakta hai.

- **SYN Flood:** attacker fake IPs se bahut saare SYN packets bhejta hai aur 3-way handshake kabhi poora nahi karta. Server ki half-open connection table bhar jaati hai aur real users ko service nahi milti.
- **Sequence number prediction / Session hijacking:** sequence numbers guess ho jaayein to attacker data inject kar sakta hai ya connection hijack kar sakta hai.
- **RST injection:** forged reset packet se valid connection kat sakta hai.
- **Port scanning:** open ports aasani se pata chal jaate hain.
- **Fix:** SYN cookies, firewall, random initial sequence numbers, rate limiting, TLS.

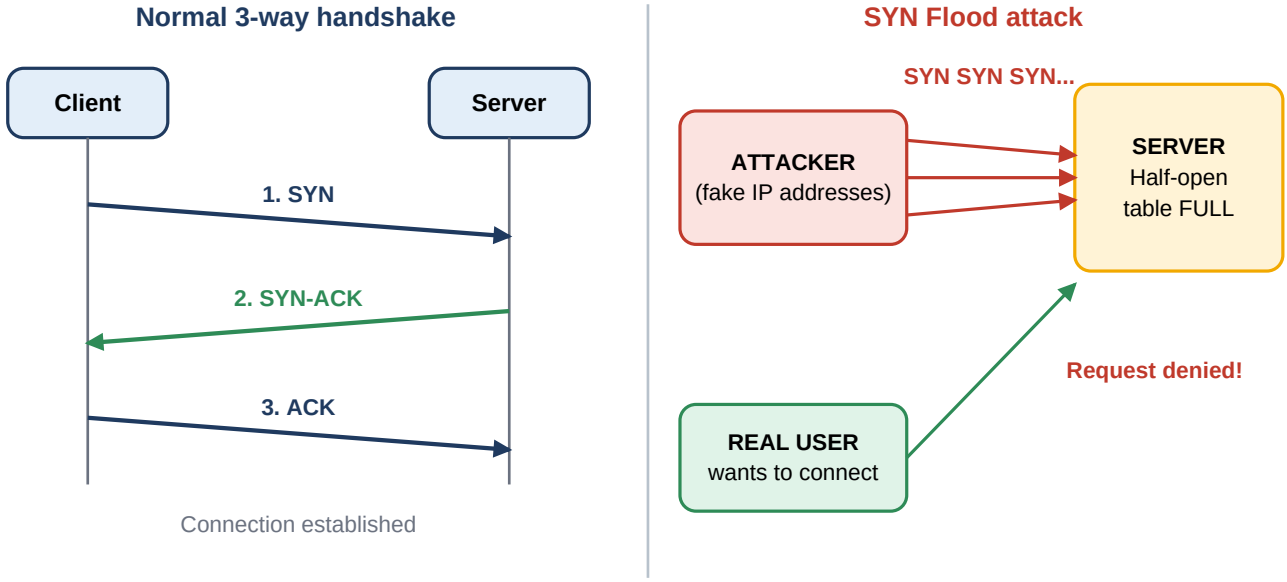


Figure 1: Normal TCP handshake and SYN Flood attack

2(c) UDP (User Datagram Protocol)

> **Real-life example**
> UDP bina receipt maange letter letterbox me daalne jaisa hai. Fast hai, par koi check nahi karta ki kisne bheja.

- **Connectionless:** handshake aur sequence number nahi hote, isliye spoofing bahut aasaan hai.
- **UDP flood:** bahut saare UDP packets server ko thaka dete hain.
- **Amplification / Reflection DDoS:** DNS/NTP servers ko chhoti spoofed request bhejne se victim ko bahut badi reply milti hai.
- Delivery guarantee aur built-in security nahi hai.
- **Fix:** unused ports block karo, rate limiting, ingress filtering, DTLS.

2(d) RIP (Routing Information Protocol)

> **Real-life example**
> Socho ek musafir har aane jaane wale se rasta poochta hai aur sab par bharosa karta hai. Jhootha insaan use galat sheher bhej sakta hai. RIP routers bhi har update par aise hi bharosa karte hain.

- RIP ek **distance-vector** protocol hai jo apni routing table neighbours se regularly share karta hai.
- **RIPv1 me authentication nahi hai.** Koi bhi fake route update bhej sakta hai.
- **Route poisoning / false routes:** traffic attacker ki taraf mud jaata hai (sniffing) ya gayab ho jaata hai (black hole).
- Updates broadcast hote hain, isliye capture aur fake karna aasaan hai.
- **Fix:** RIPv2 with MD5 authentication, passive interfaces, route filters.

2(e) OSPF (Open Shortest Path First)

- OSPF ek **link-state** protocol hai. Routers network ka map banane ke liye **LSAs** (link state advertisements) exchange karte hain.
- Authentication configure na ho (default me none hota hai) to attacker **fake LSAs** daal kar poora map badal sakta hai.
- Results: traffic redirect (MITM), black hole, routing loops, DoS.
- Hello packets spoof karke fake neighbours banaye ja sakte hain.
- **Fix:** OSPF authentication (MD5/SHA), passive interfaces, area design, route filtering.

2(f) HTTP (HyperText Transfer Protocol)

> **Real-life example**
> HTTP postcard bhejne jaisa hai. Jo bhi use handle karega padh sakta hai. Postcard par likha password sabko dikhta hai.

- **Encryption nahi:** username, password aur cookies plain text me jaate hain aur sniff ho sakte hain.
- **Server authentication nahi:** pakka nahi hota ki website asli hai, isliye **MITM** aur fake sites possible hain.
- **Session hijacking:** chori hui cookies se attacker tumhare naam se login kar sakta hai.
- **Data tampering:** content raste me badla ja sakta hai.
- **Fix: HTTPS**, secure cookies, HSTS, input validation use karo.

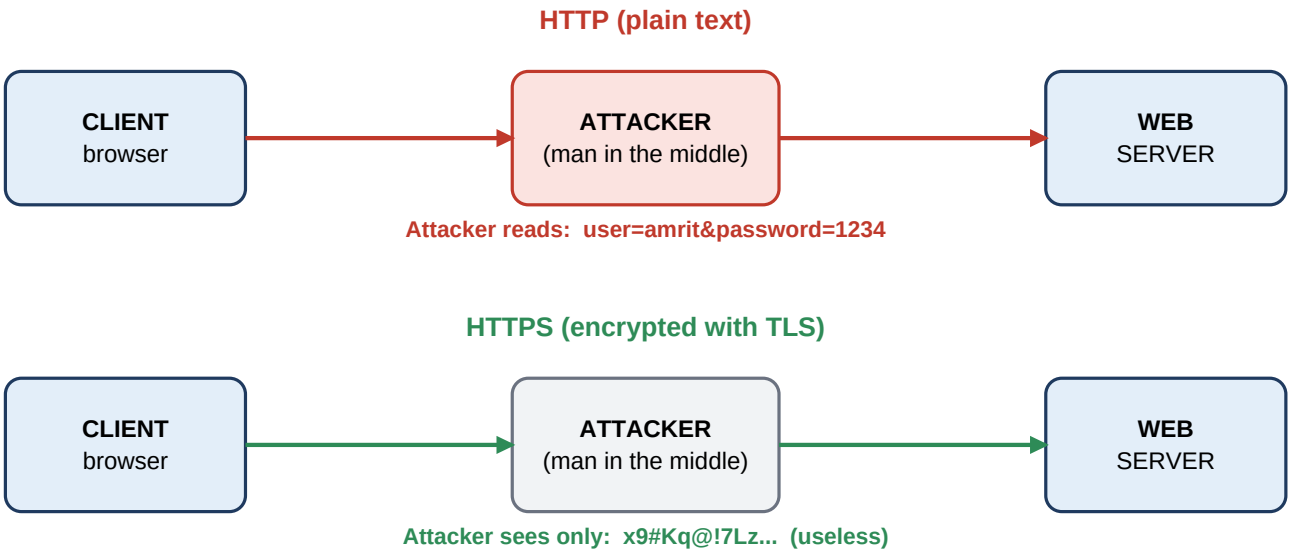


Figure 2: HTTP vs HTTPS when an attacker is in the middle

2(g) SMTP (Simple Mail Transfer Protocol)

> **Real-life example**
> Email ke 'From' field me koi bhi koi bhi naam likh sakta hai, jaise courier parcel par fake naam likhna. SMTP use check nahi karta.

- Default me **sender authentication nahi**: **email spoofing**, phishing aur spam aasaan hain.
- **Plain text:** emails aur passwords sniff ho sakte hain.
- **Open relay:** galat configure server kisi ka bhi mail forward kar deta hai, spammers iska misuse karte hain.
- **Information leak:** VRFY aur EXPN commands valid user names bata dete hain.
- **Fix:** STARTTLS/SMTPS, SMTP AUTH, SPF, DKIM, DMARC, open relay band karo, spam filters.

Summary table: protocol weaknesses

Protocol	Main weakness	Attack	Solution
IP	Authentication nahi	IP spoofing, Smurf	IPSec, ingress filtering
TCP	Handshake ka misuse	SYN flood, hijacking	SYN cookies, firewall
UDP	Connectionless	UDP flood, amplification	Rate limit, block ports
RIP	Authentication nahi (v1)	Fake route update	RIPv2 + MD5
OSPF	Fake LSAs accept hote hain	Route injection	OSPF authentication
HTTP	Plain text	Sniffing, MITM	HTTPS (TLS)
SMTP	Sender check nahi	Email spoofing, spam	SPF, DKIM, STARTTLS

Topic 3: Security Weaknesses in Networking Devices

3(a) Hub

> **Real-life example**
> Hub kamre me chillakar message bolne jaisa hai. Kamre me sab sunte hain, chahe message ek insaan ke liye ho.

- Hub Layer 1 par kaam karta hai aur **har packet sabhi ports par broadcast** karta hai.
- Hub se juda koi bhi PC sniffer (jaise Wireshark) se sabka data padh sakta hai.
- Filtering, management aur security features nahi. Ek hi shared collision domain.
- **Fix:** hub ki jagah switch lagao.

3(b) Switch

> **Real-life example**
> Switch post office ke clerk jaisa hai jo har letter sahi ghar tak pahunchata hai, addresses ki list (MAC table) dekh kar. Attackers is list ko confuse karne ki koshish karte hain.

- **MAC flooding:** hazaron fake MAC addresses switch ki CAM table bhar dete hain. Phir switch hub jaisa broadcast karne lagta hai aur attacker sniff kar sakta hai.
- **ARP spoofing / poisoning:** attacker fake ARP replies bhejta hai jisse traffic uske through jaata hai (MITM).
- **VLAN hopping:** attacker switch spoofing ya double tagging se doosre VLAN tak pahunch jaata hai.
- **STP attack:** fake BPDU se attacker root bridge ban jaata hai.
- **DHCP starvation / spoofing** aur default passwords.
- **Fix:** port security, Dynamic ARP Inspection, DHCP snooping, BPDU guard, unused ports band karo, default passwords badlo.

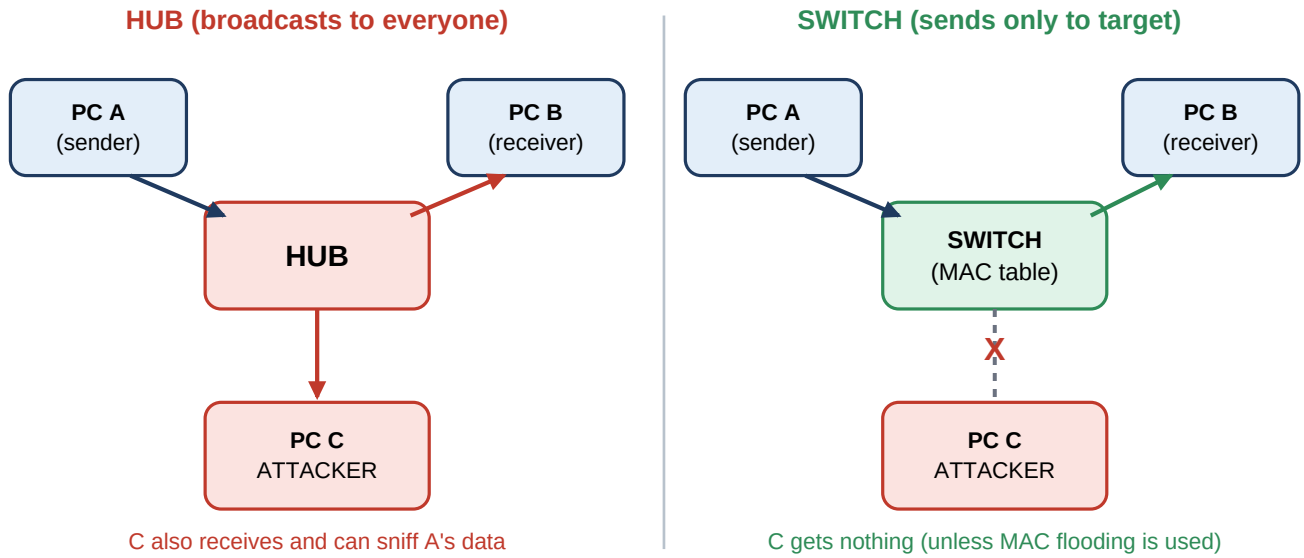


Figure 3: Hub vs Switch (sniffing)

3(c) Router

> **Real-life example**
> Router tumhare network ka main gate hai. Gate ka lock kamzor ho ya key 'admin/admin' ho to chor aasani se ghus jaate hain.

- Admin panel par **default / weak passwords**.
- **Insecure management:** Telnet aur HTTP management passwords plain text me bhejte hain.
- **Purana firmware** jisme known vulnerabilities hoti hain.
- **Galat ACLs** aur unnecessary open services.
- **Routing attacks** (fake updates), source routing, ICMP redirects aur DoS floods.
- **Fix:** strong passwords, Telnet ki jagah SSH, firmware update, ACLs, unused services band karo, routing authentication.

3(d) WiFi (Wireless Network)

> **Real-life example**
> WiFi data hawa me bhejta hai. Ye loudspeaker par bolne jaisa hai: aas paas khada koi bhi bina kisi wire ke sun sakta hai.

- **Eavesdropping:** radio signals deewaron se bhi nikalte hain, isliye bahar wale traffic capture kar sakte hain.
- **Weak encryption:** WEP kuch minutes me tut jaata hai. Weak password wala WPA/WPA2 dictionary attack se crack ho sakta hai.
- **Rogue AP / Evil Twin:** same naam ka fake access point users ko connect karwa leta hai.
- **WPS PIN** weakness aur router ka **default admin password**.
- **Deauthentication attack:** attacker users ko disconnect karke reconnect handshake capture karta hai.
- **Fix:** WPA3 ya WPA2-AES, lamba passphrase, WPS band, default passwords badlo, firmware update, WPA2-Enterprise (802.1X), alag guest network.

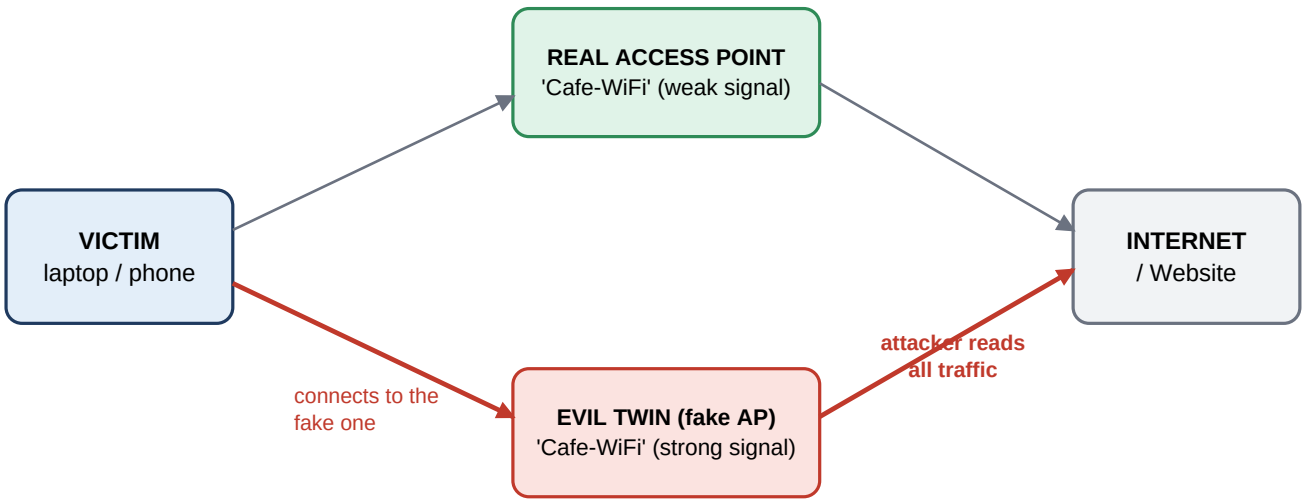


Figure 4: Evil Twin attack on WiFi

Topic 4: Security Solutions

Ab problems pata hain, to ye hain main solutions: protocols ke liye (IPSec, HTTPS) aur devices/networks ke liye (VLAN, VPN, Ingress Filtering).

4(a) IPSec (IP Security)

Real-life example

Normal IP postcard hai. IPSec letter ko **sealed, signed envelope** me daalta hai: koi padh, badal ya sender fake nahi kar sakta.

Definition

IPSec protocols ka ek suite hai jo **network layer (Layer 3)** par IP communication ko secure karta hai: confidentiality, integrity, authentication aur anti-replay protection deta hai.

- **AH (Authentication Header):** authentication aur integrity deta hai, par **encryption nahi**.
- **ESP (Encapsulating Security Payload): encryption** + authentication + integrity deta hai.
- **IKE (Internet Key Exchange):** keys aur **Security Association (SA)** set karta hai.
- **Transport mode:** sirf data protect hota hai (host to host).
- **Tunnel mode:** poora original packet protect hokar naye IP header me wrap hota hai (gateway to gateway, VPN me use hota hai).

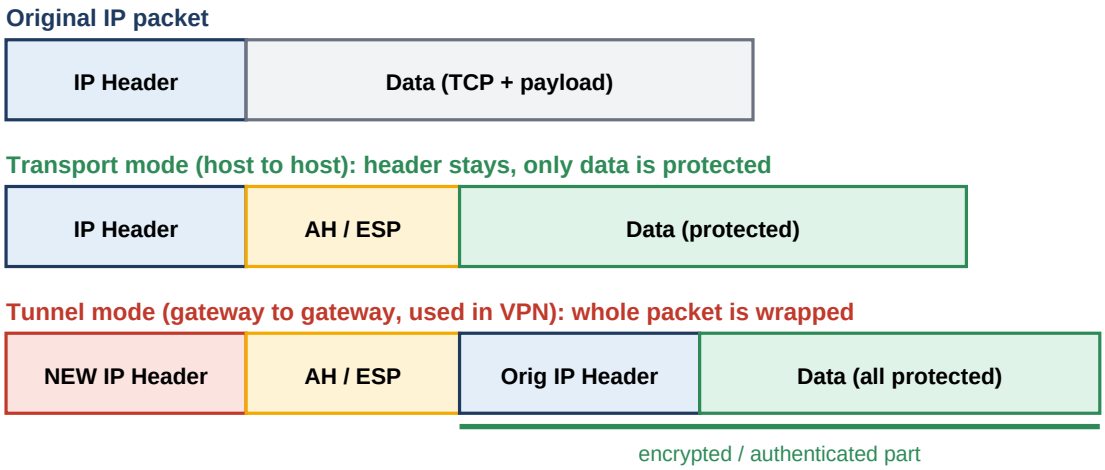


Figure 5: IPSec transport and tunnel modes

4(b) HTTPS (HTTP Secure)

Real-life example

HTTP postcard hai; HTTPS ek **locked box** hai jise sirf website khol sakti hai. Browser ka padlock dikhata hai ki website ki pehchan check hui hai.

Definition

HTTPS HTTP hai jo **SSL/TLS** ke upar chalta hai. Ye port **443** aur digital certificate use karke encryption, server authentication aur integrity deta hai.

- **Step 1:** browser hello bolta hai aur supported TLS versions bhejta hai.
- **Step 2:** server apna **digital certificate** (CA ne diya hua) bhejta hai.
- **Step 3:** browser certificate verify karta hai.
- **Step 4:** asymmetric cryptography se **session key** safely share hoti hai.
- **Step 5:** baaki sara data tez **symmetric** session key se encrypt hota hai.

4(c) VLAN (Virtual LAN)

Real-life example

Ek bade office hall ko partitions se baanta ja sakta hai: Accounts, Students, Guests. Ek cabin ke log doosron ko nahi sun sakte. VLAN yahi kaam switch ke andar karta hai.

Definition

VLAN switch par devices ka logical group hai jo alag network ki tarah kaam karta hai, chahe unki physical location kuch bhi ho. Ye **IEEE 802.1Q** tagging use karta hai.

- Chhota **broadcast domain**, isliye sniff karne layak traffic kam.
- Sensitive departments aur guests ko ek doosre se **alag** karta hai.
- Inter-VLAN traffic router ya Layer-3 switch se guzarta hai, jahan ACL/firewall lag sakte hain.
- VLAN ko protect karo: DTP band karo, native VLAN badlo, unused ports shut karo.

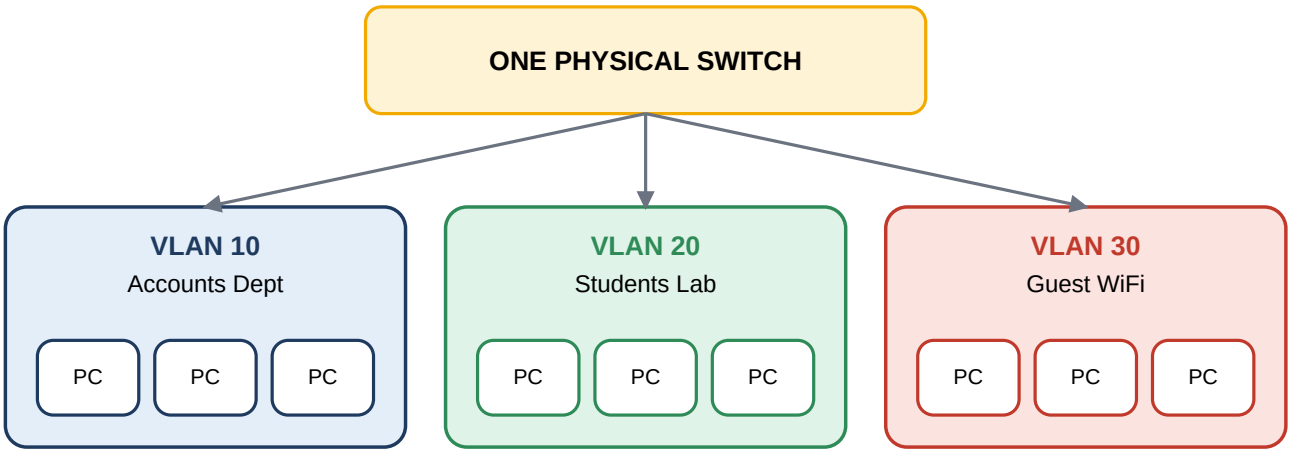


Figure 6: VLAN segmentation

4(d) VPN (Virtual Private Network)

Real-life example

VPN public sadak ke beech banaya gaya ek **private, dhaka hua tunnel** hai. Tunnel ke andar ki gadiyan bahar wale na dekh sakte hain na rok sakte hain.

Definition

VPN internet jaise public network par ek secure, encrypted tunnel banata hai, jisse remote users ya offices private network se safely connect ho sakte hain.

- **Types:** Remote-access VPN (employee se office) aur Site-to-site VPN (office se office).
- **Protocols:** IPSec, SSL/TLS VPN, OpenVPN, L2TP. (PPTP purana aur weak hai.)
- **Confidentiality, integrity aur authentication** deta hai.
- Public WiFi par users ko protect karta hai aur data sniffers se chhupata hai.

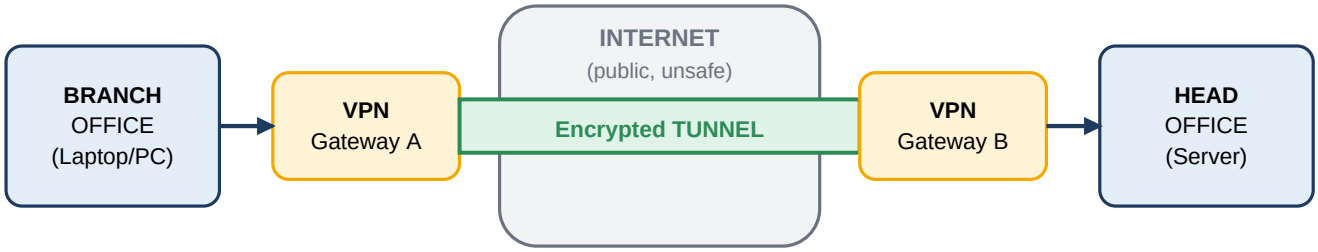


Figure 7: VPN tunnel

4(e) Ingress Filtering

Real-life example

Main gate par guard us ajnabi ko rokta hai jo bahar se aakar bolta hai 'main is ghar ka malik hu'. Ingress filtering IP addresses ke liye wahi guard hai.

Definition

Ingress filtering (BCP 38 / RFC 2827) ek technique hai jisme edge router un **incoming packets ko drop kar deta hai jinka source IP address** us network ya interface ke liye valid nahi hota.

- Ye **IP spoofing** rokta hai aur DDoS (reflection attacks sahit) kam karta hai.
- Example: internet se aaya packet jo private/internal address (10.x.x.x) claim kare, drop ho jaata hai.
- **Egress filtering** ulta hai: outgoing packets jinka source IP tumhare network ka nahi hai, block hote hain.
- Routers par ACLs ya **uRPF** (unicast Reverse Path Forwarding) se implement hota hai.

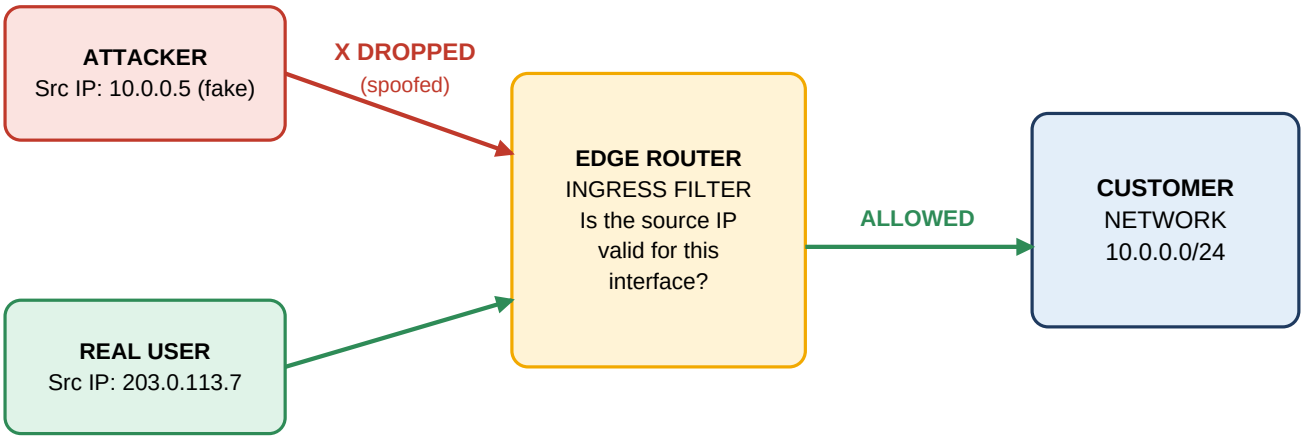


Figure 8: Ingress filtering at the edge router

PART B: Important Questions aur Model Answers (Hinglish)

Priority: **Very Likely** = BTEUP me sabse zyada aata hai. **Likely** = achha chance. **Possible** = aa sakta hai. Har answer exam format me hai: **Definition, Points, Diagram/Table, Conclusion**.

Q1. IP, TCP aur UDP ki security weaknesses explain karo. — VERY LIKELY

Definition / Parichay: IP, TCP aur UDP internet ke core protocols hain jo bina security ke design hue the. Inki main weaknesses neeche hain.

Points

- **IP:** authentication nahi, isliye source address fake ho sakta hai (IP spoofing). Encryption nahi, isliye packets sniff ho sakte hain. Fragmentation attacks (Teardrop, Ping of Death) aur ICMP attacks (Smurf, ping flood) possible hain.
- **TCP:** 3-way handshake ka SYN flood se misuse hota hai. Predictable sequence numbers se session hijacking hoti hai. Forged RST packet connection tod sakta hai. Port scanning aasaan hai.
- **UDP:** connectionless hai, handshake nahi, isliye spoofing bahut aasaan hai. UDP flood aur amplification/reflection DDoS (DNS, NTP) common hain. Delivery guarantee nahi.
- **Solutions:** IPSec, ingress/egress filtering, firewalls, SYN cookies, rate limiting, unused ports block karna.

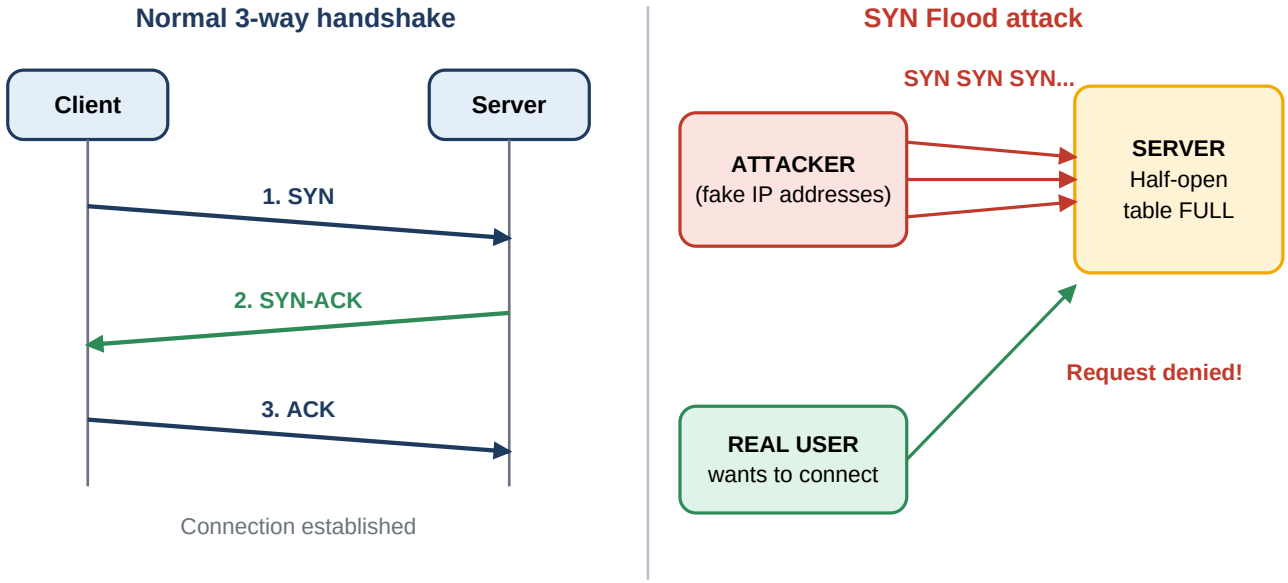


Figure: TCP handshake and SYN flood

Protocol	Weakness	Attack
IP	Authentication / encryption nahi	IP spoofing, Smurf, Teardrop
TCP	Handshake misuse, predictable sequence	SYN flood, hijacking, RST injection
UDP	Connectionless	UDP flood, amplification

Conclusion: In protocols ko safely use karne ke liye IPSec, firewalls aur filtering jaisi extra security zaroori hai.

Q2. RIP aur OSPF routing protocols ki security weaknesses explain karo. LIKELY

Definition / Parichay: RIP (distance-vector) aur OSPF (link-state) routing protocols hain jo routers ke beech route information share karte hain. Ye mile hue updates par bharosa karte hain.

Points

- **RIP:** RIPv1 me authentication nahi hai, isliye koi bhi false route updates bhej sakta hai.
- RIP updates regularly broadcast hote hain, isliye capture aur forge karna aasaan hai.
- **OSPF:** authentication configure na ho to fake LSAs aur fake Hello packets accept ho jaate hain.
- **Effects:** route poisoning, traffic attacker ki taraf redirect (MITM), black hole, routing loops aur DoS.
- **Solutions:** RIPv2 / OSPF authentication (MD5 ya SHA), passive interfaces, route filtering aur secure router management.

Point	RIP	OSPF
Type	Distance-vector	Link-state
Default security	Authentication nahi (v1)	Default me authentication nahi
Attack	Fake route update	Fake LSA injection
Fix	RIPv2 + MD5	OSPF MD5/SHA authentication

Conclusion: Attackers ko traffic path control karne se rokne ke liye routing updates ki authentication zaroori hai.

Q3. HTTP aur SMTP ki security weaknesses explain karo. Inhe secure kaise kiya ja sakta hai? VERY LIKELY

Definition / Parichay: HTTP web browsing ke liye aur SMTP email bhejne ke liye use hota hai. Dono default me data plain text me bhejte hain.

Points

- **HTTP weaknesses:** encryption nahi (passwords aur cookies sniff ho sakte hain), server authentication nahi (MITM aur fake sites), session hijacking aur data tampering.
- **SMTP weaknesses:** sender authentication nahi (email spoofing, phishing, spam), plain text, open relay misuse aur VRFY/EXPN se information leak.
- **HTTP secure karna:** HTTPS (TLS), secure cookies, HSTS aur input validation.
- **SMTP secure karna:** STARTTLS/SMTPS, SMTP AUTH, SPF, DKIM, DMARC, open relays band karna aur spam filters.

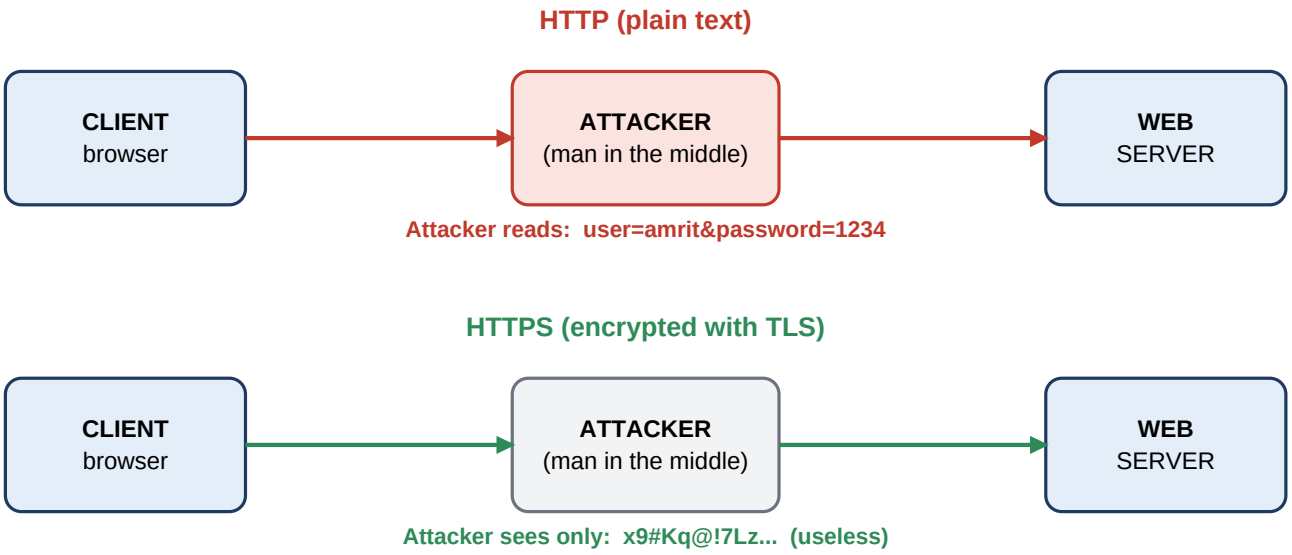


Figure: HTTP vs HTTPS under a man-in-the-middle attack

Conclusion: Encrypted versions (HTTPS, SMTP over TLS) authentication ke saath use karne se zyada tar weaknesses khatam ho jaati hain.

Q4. Hub, Switch aur Router ki security weaknesses explain karo. VERY LIKELY

Definition / Parichay: Hub, switch aur router network ke basic devices hain. Har ek ki apni security weaknesses hain.

Points

- **Hub:** har packet sabhi ports par broadcast karta hai, isliye connected koi bhi machine data sniff kar sakti hai. Filtering ya management nahi hai.
- **Switch:** MAC flooding, ARP spoofing, VLAN hopping, STP attacks, DHCP starvation aur default passwords.
- **Router:** default/weak passwords, Telnet management, purana firmware, galat ACLs, routing protocol attacks aur DoS.
- **Solutions:** hub ki jagah switch, port security, DAI, DHCP snooping, strong passwords, SSH, firmware update, ACLs.

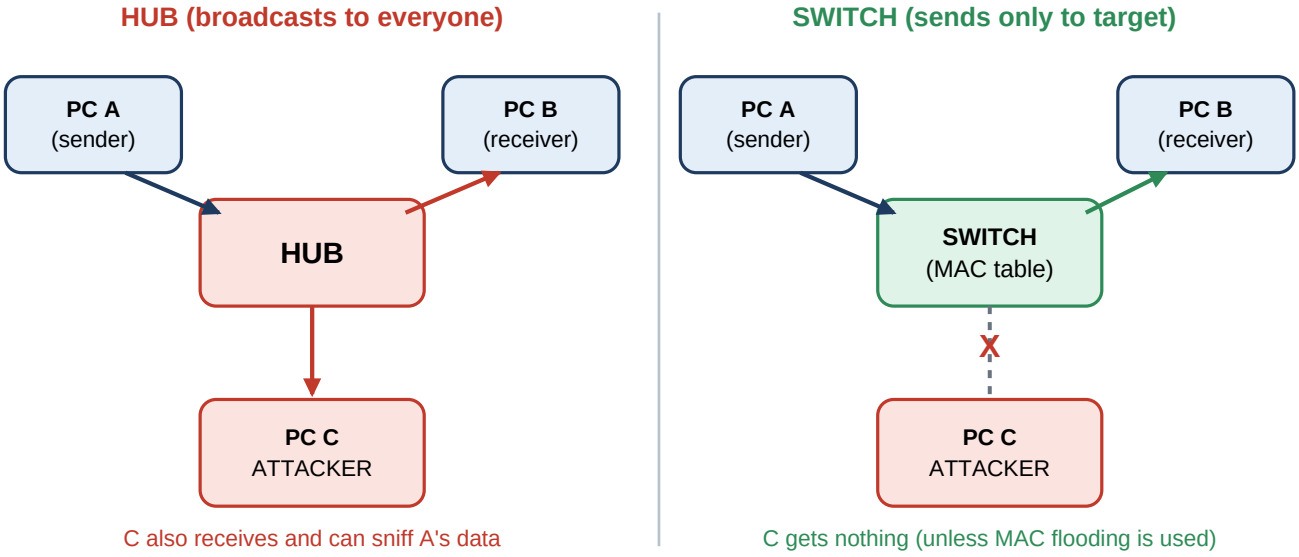


Figure: Hub broadcasts to all, switch sends only to the target

Device	Layer	Key weakness	Fix
Hub	1	Broadcast, sniffing	Switch use karo
Switch	2	MAC flooding, ARP spoofing	Port security, DAI
Router	3	Weak config, routing attacks	ACL, SSH, updates

Conclusion: Sahi configuration aur purane devices badalne se network-level attacks bahut kam ho jaate hain.

Q5. WiFi ki security weaknesses explain karo aur wireless network ko secure kaise karein. VERY LIKELY

Definition / Parichay: WiFi ek wireless LAN hai jo radio waves se data bhejta hai. Medium khuli hawa hai, isliye wired network se isse attack karna aasaan hai.

Points

- **Eavesdropping:** signals building ke bahar se capture ho sakte hain.
- **Weak encryption:** WEP aasani se tut jaata hai; weak password wala WPA/WPA2 crack ho sakta hai.
- **Rogue AP / Evil Twin** aur **MITM** attacks.
- **WPS PIN** weakness, default admin passwords aur deauthentication attacks.
- **Security measures:** WPA3 ya WPA2-AES, strong lamba passphrase, WPS band, default passwords badlo, firmware update, WPA2-Enterprise (802.1X), alag guest network, public WiFi par VPN.

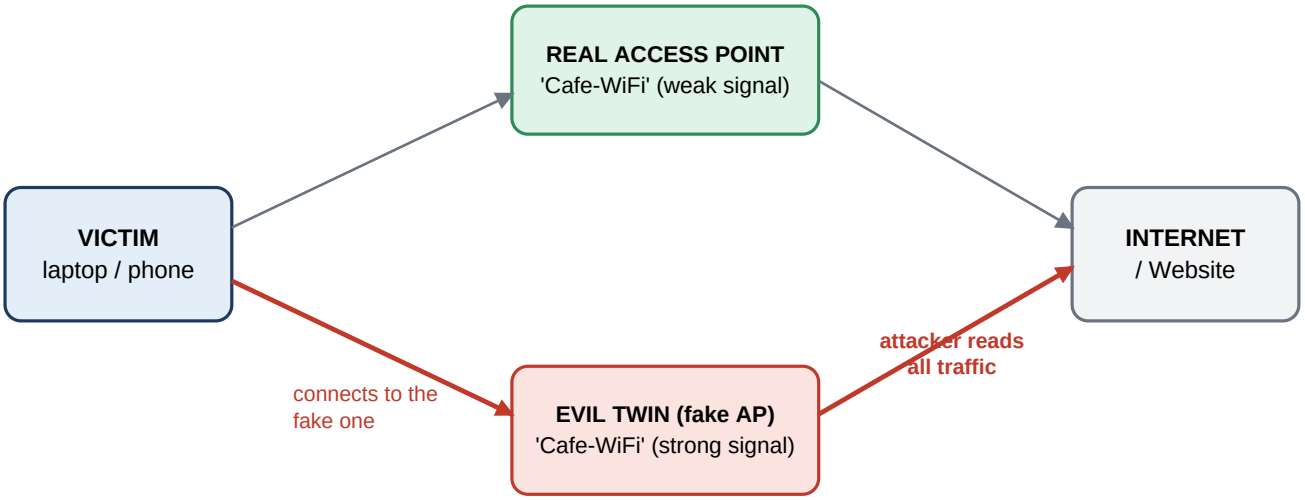


Figure: Evil Twin attack

Standard	Security level
WEP	Bahut weak, tuta hua
WPA	Weak, purana
WPA2 (AES)	Achha
WPA3	Sabse achha (current)

Conclusion: Strong encryption standard aur achhe password management se wireless network safe rehta hai.

Q6. IPSec kya hai? Uske components aur modes explain karo. VERY LIKELY

Definition / Parichay: IPSec (IP Security) protocols ka suite hai jo network layer par IP communication ko secure karta hai: confidentiality, integrity, authentication aur anti-replay protection deta hai.

Points

- **AH (Authentication Header):** authentication aur integrity, encryption nahi.
- **ESP (Encapsulating Security Payload):** encryption, authentication aur integrity.
- **IKE (Internet Key Exchange):** keys negotiate karta hai aur Security Association (SA) banata hai.
- **Transport mode:** sirf payload protect hota hai; do hosts ke beech use hota hai.
- **Tunnel mode:** poora original packet protect hota hai aur naya IP header judta hai; gateways ke beech (VPN) use hota hai.

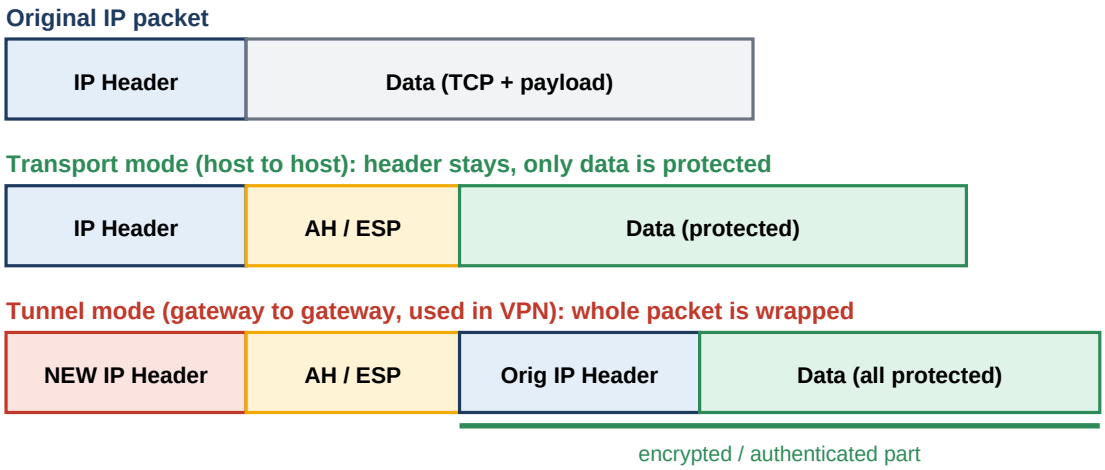


Figure: IPSec transport and tunnel modes

Point	AH	ESP
Encryption	Nahi	Haan
Authentication / Integrity	Haan (poora packet)	Haan (payload)
Use	Jab sirf authenticity chahiye	Jab secrecy bhi chahiye

Conclusion: IPSec IP traffic secure karne ka standard tarika hai aur zyada tar VPNs ka base hai.

Q7. HTTPS kya hai? Ye kaise kaam karta hai? LIKELY

Definition / Parichay: HTTPS (HTTP Secure) HTTP hai jo port 443 par SSL/TLS ke upar chalta hai. Ye encryption, server authentication aur data integrity deta hai.

Points

- Browser supported TLS versions aur ciphers ke saath hello bhejta hai.
- Server Certificate Authority (CA) ka diya hua apna **digital certificate** bhejta hai.
- Browser certificate verify karta hai (validity, domain, trusted CA).
- Asymmetric cryptography se **session key** safely exchange hoti hai.
- Aage ka data symmetric session key se encrypt hota hai aur hashes integrity pakki karte hain.
- **Fayde:** sniffing, MITM aur tampering rokta hai. Browser me padlock icon dikhta hai.

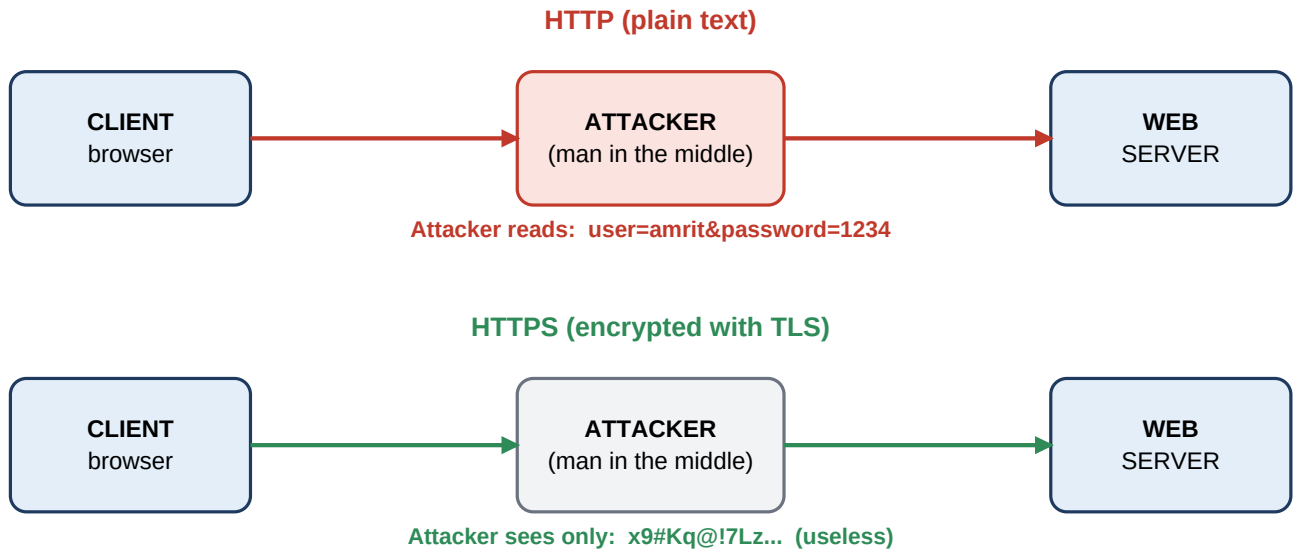


Figure: HTTPS protects data from a man-in-the-middle

Conclusion: Jis website par password, payment ya personal data ho, wahan HTTPS zaroori hai.

Q8. VLAN kya hai? Uske advantages explain karo. LIKELY

Definition / Parichay: VLAN (Virtual LAN) ek ya zyada switches par devices ka logical group hai jo physical location se alag ek separate network ki tarah kaam karta hai. Ye IEEE 802.1Q tagging use karta hai.

Points

- **Security:** departments aur guests alag ho jaate hain, isliye sniffing aur attacks limited rehte hain.
- **Chhota broadcast domain:** kam traffic aur behtar performance.
- **Flexibility:** users ko location nahi, department ke hisab se group kar sakte hain.
- **Control:** VLANs ke beech communication router ya Layer-3 switch se guzarti hai, jahan ACLs lag sakte hain.
- **Risk:** VLAN hopping; DTP band karke, native VLAN badal kar aur unused ports shut karke roka jaata hai.

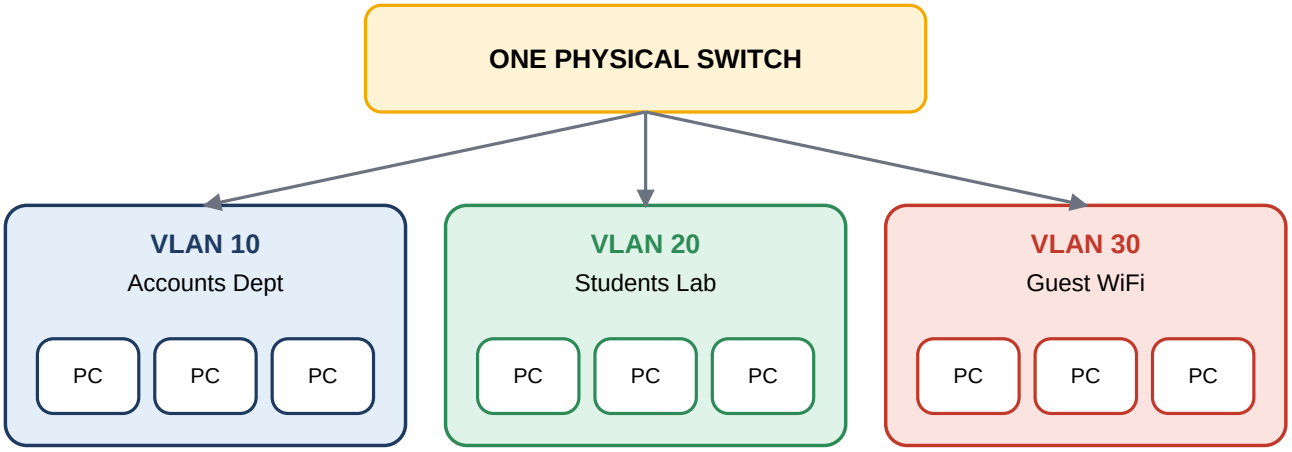


Figure: VLAN segmentation

Conclusion: VLANs LAN ki security aur performance dono badhate hain.

Q9. VPN kya hai? Uske types aur working explain karo. LIKELY

Definition / Parichay: VPN (Virtual Private Network) internet jaise public network par encrypted tunnel banata hai, jisse private aur secure communication milti hai.

Points

- **Working:** VPN client/gateway user ko authenticate karta hai, data encrypt karke tunnel se VPN server ko bhejta hai, jo use decrypt karta hai.
- **Remote-access VPN:** ek user office network se connect hota hai.
- **Site-to-site VPN:** do office networks safely jud jaate hain.
- **Protocols:** IPSec, SSL/TLS, OpenVPN, L2TP.
- **Fayde:** confidentiality, integrity, authentication, public WiFi par safety, leased line se kam kharch.

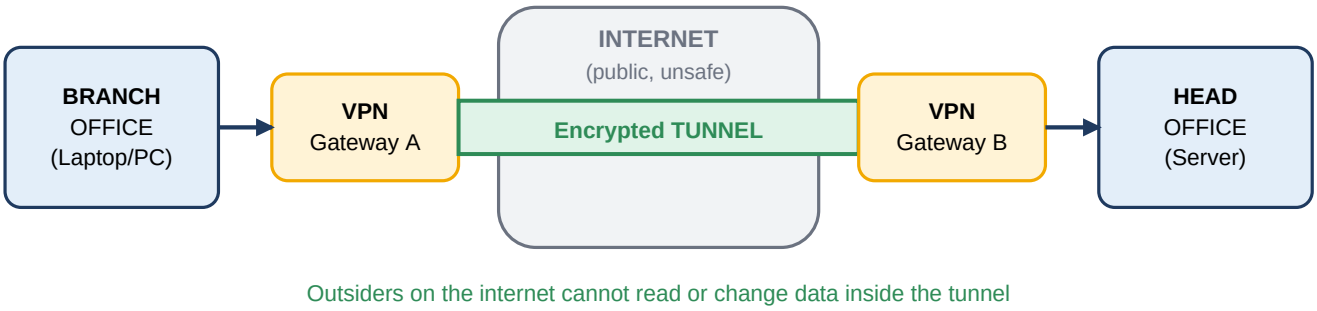


Figure: VPN tunnel across the internet

Conclusion: VPN insecure network par bhi private resources tak secure access deta hai.

Q10. Ingress Filtering kya hai? Ye kyun zaroori hai? LIKELY

Definition / Parichay: Ingress filtering (BCP 38 / RFC 2827) ek network security technique hai jisme router un incoming packets ko drop karta hai jinka source IP address us interface ke liye valid nahi hota.

Points

- Ye **IP spoofing** rokta hai, jisme attackers fake source address se apni pehchan chhupate hain.
- Ye **DDoS** aur reflection/amplification attacks (Smurf, DNS amplification) kam karta hai.
- **Example:** internet se aaya packet jiska source 10.0.0.5 (private address) hai, drop ho jaata hai.
- **Egress filtering** iska ulta hai aur aise outgoing packets block karta hai jinka source address tumhare network ka nahi hai.
- **Implementation:** edge routers par ACLs ya uRPF (Unicast Reverse Path Forwarding).

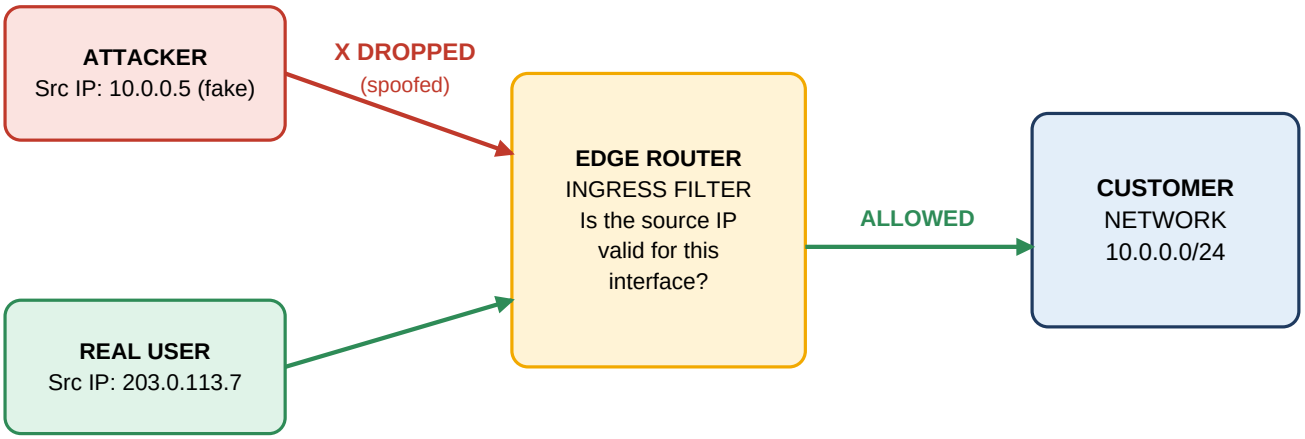


Figure: Ingress filtering at the edge router

Conclusion: Agar ISPs aur organizations ingress filtering lagayein to spoofed-source attacks bahut mushkil ho jaate hain.

Q11. SYN Flood attack explain karo. POSSIBLE

Definition / Parichay: SYN flood ek Denial-of-Service attack hai jo TCP 3-way handshake ka misuse karke server ke resources khatam kar deta hai.

Points

- Normal handshake: client SYN bhejta hai, server SYN-ACK bhejta hai, client ACK bhejta hai.
- Attack me attacker bahut saare SYN packets bhejta hai, aksar fake source IPs se, aur last ACK kabhi nahi bhejta.
- Server bahut saare **half-open connections** rakhta hai aur uski backlog queue bhar jaati hai.
- Real users connect nahi kar paate, isliye service deny ho jaati hai.
- **Prevention:** SYN cookies, SYN timeout kam karna, backlog badhana, firewall/IPS rate limiting, ingress filtering.

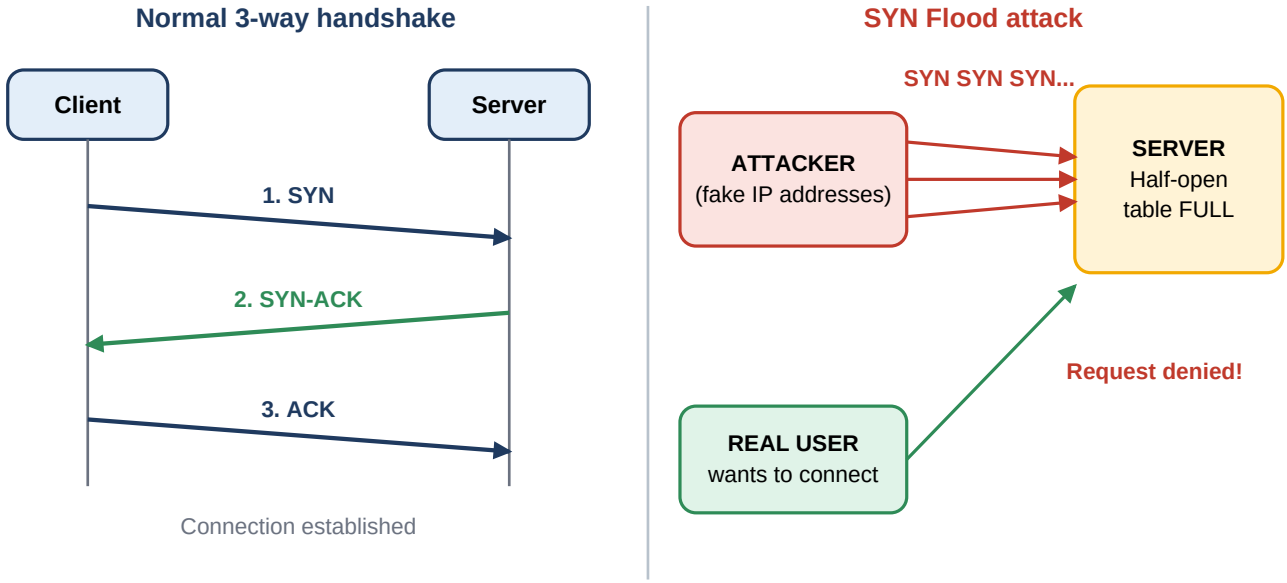


Figure: Normal handshake and SYN flood

Conclusion: SYN flood simple par powerful attack hai, isliye servers par SYN cookies aur rate limiting lagana zaroori hai.

Q12. Hub aur Switch me difference likho. MAC flooding aur ARP spoofing explain karo. — POSSIBLE

Definition / Parichay: Hub ek simple Layer-1 device hai jo data sabhi ports par repeat karta hai; switch Layer-2 device hai jo MAC address table se data sirf destination port par bhejta hai.

Points

- **MAC flooding:** attacker switch ko fake MAC addresses se bhar deta hai jab tak CAM table full na ho jaaye. Phir switch hub jaisa kaam karta hai aur attacker sniff kar sakta hai. **Fix:** port security.
- **ARP spoofing:** attacker fake ARP replies bhejta hai jisse uska MAC address kisi aur device ke IP se jud jaata hai, aur traffic uske through jaata hai (MITM). **Fix:** Dynamic ARP Inspection, static ARP entries.

Point	Hub	Switch
Layer	Layer 1 (Physical)	Layer 2 (Data link)
Data sending	Sabhi ports par broadcast	Sirf destination port par
Intelligence	Koi nahi	MAC address table use karta hai
Security	Bahut kam, sniffing aasaan	Behtar, par MAC flooding / ARP spoofing possible
Collision domain	Sabke liye ek	Har port ka alag

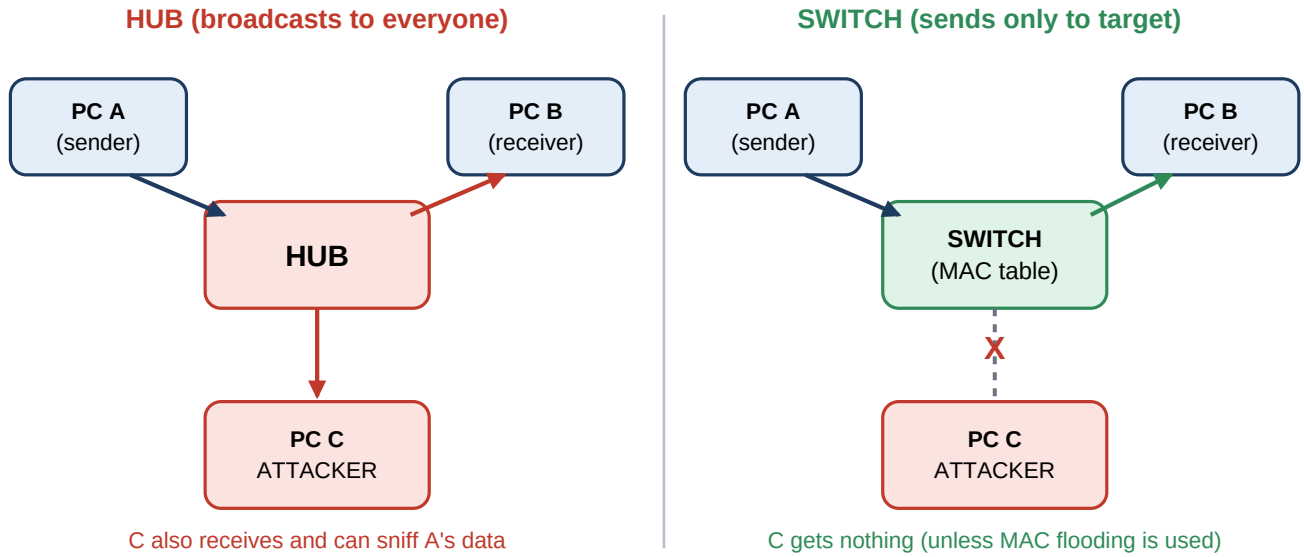


Figure: Hub vs Switch

Conclusion: Switch hub se zyada secure hai, par use bhi MAC flooding aur ARP spoofing se bachana padta hai.

Quick Revision Sheet (exam se pehle raat ko padho)

Term	Ye yaad rakho
IP spoofing	Fake source IP. Fix: ingress filtering, IPSec
SYN flood	Bahut SYN, last ACK nahi, half-open table full. Fix: SYN cookies
Smurf	Fake source ke saath broadcast ping (ICMP)
UDP amplification	Chhoti spoofed request, victim ko badi reply (DNS, NTP)
RIP	v1 me authentication nahi. Fix: RIPv2 + MD5
OSPF	Fake LSA injection. Fix: OSPF authentication
HTTP vs HTTPS	Plain text vs TLS encrypted (port 80 vs 443)
SMTP	Sender check nahi. Fix: SPF, DKIM, DMARC, STARTTLS
Hub	Layer 1, broadcast, sniffing aasaan
Switch attacks	MAC flooding, ARP spoofing, VLAN hopping, STP, DHCP
Switch fixes	Port security, DAI, DHCP snooping, BPDU guard
Router	Default password, Telnet, purana firmware. Fix: SSH, ACL, updates
WiFi	WEP tuta hua. WPA3 / WPA2-AES use karo. Evil twin, WPS
IPSec	AH (encryption nahi), ESP (encryption), IKE (keys). Transport / Tunnel
HTTPS	HTTP + TLS, port 443, CA ka certificate
VLAN	Logical LAN split, IEEE 802.1Q, chhota broadcast domain
VPN	Encrypted tunnel. Remote-access aur Site-to-site
Ingress filtering	Invalid source IP wale incoming packets drop (BCP 38)

Yaad rakhne ke tricks

Protocol weakness: 'No A-E-I' yaad rakho = No Authentication, No Encryption, No Integrity.
Switch attacks: M-A-V-S-D = MAC flooding, ARP spoofing, VLAN hopping, STP, DHCP.
IPSec: AH = sirf Authenticate, ESP = Encrypt bhi.
Ingress vs Egress: Ingress = **andar** aane wala, Egress = **bahar** jaane wala.

Common Mistakes jo Students Karte Hain

- **AH** aur **ESP** ko mix karna. Sirf ESP encrypt karta hai.
- HTTPS aur HTTP ko alag protocols likhna. HTTPS **HTTP over TLS/SSL** hai.
- **Ingress** (incoming) aur **egress** (outgoing) filtering ko mix karna.
- Switch ko bilkul safe bolna. Us par bhi MAC flooding aur ARP spoofing ho sakti hai.
- Weakness likhne ke baad **solution** likhna bhool jana. Hamesha fix bhi likho.
- Diagrams na banana (SYN flood, IPSec modes, VPN, VLAN).

Self-Test (bina dekhe likho, har question 5 marks)

- SYN flood attack diagram ke saath explain karo.
- HTTP aur SMTP ki weaknesses likho, har ek ka ek solution do.
- MAC flooding aur ARP spoofing explain karo.
- AH aur ESP me difference likho. Transport aur tunnel modes kya hain?
- Ingress filtering kya hai? Example do.

Self-Test Answers (likhne ke baad check karo)

1. Attacker fake IPs se bahut SYN packets bhejta hai aur last ACK nahi bhejta. Server ki half-open table bhar jaati hai aur real users deny ho jaate hain. Fix: SYN cookies, rate limiting. SYN flood figure banao.

2. HTTP: plain text, sniffing aur MITM, fix HTTPS. SMTP: sender authentication nahi (email spoofing), plain text, fix SPF/DKIM ya STARTTLS.

3. MAC flooding switch ki CAM table fake MACs se bhar deta hai jisse wo hub jaisa ho jaata hai (fix: port security). ARP spoofing fake ARP replies bhejta hai jisse traffic attacker ke through jaata hai (fix: DAI).

4. AH authentication aur integrity deta hai par encryption nahi; ESP encryption bhi deta hai. Transport mode sirf payload protect karta hai (host to host); tunnel mode poora packet protect karke naya IP header lagata hai (gateway to gateway, VPN).

5. Edge router un incoming packets ko drop karta hai jinka source IP us interface ke liye valid nahi, isse IP spoofing rukti hai. Example: internet se aaya packet jiska source 10.0.0.5 (private address) hai, drop ho jaata hai.

Unit 2 complete!
Agar 5 me se 4 ya zyada sahi hue to **Unit 3** par jao. Nahi to Part A me jo topics miss hue unhe dobara padho.